Christos Hadjinikolis

Engineering Manager | Production ML, Streaming & Agent Infrastructure
Ph.D. Computer Science | Product Engineering, AI Governance & Developer Tooling
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profiles
website
LinkedIn
GitHub
Google Scholar

leadership
6 direct reports
10-person team
hiring/coaching
product partnership

platform
backend APIs
model serving
Kafka/Flink
event-driven systems
AWS/GCP

AI agents
MCP connectors
tool routing
approval gates
traces/replay
AI-assisted dev

quality
CI/CD
observability
runbooks/rollback
DORA/Jira signals
MTTR under 30 min

open source
PyPI: dynamicio
skeleton-replay
JetBrains plugin

about me

Engineering Manager with **16 years** across software engineering, production ML, streaming/event-driven systems, consulting, and AI research. At Vortexa I manage **6** direct reports and lead a **10-person** cross-functional team responsible for live backend systems where model quality, product trust, reliability, and delivery discipline matter every day. I remain technically involved in architecture and code review across **Kafka/Flink**, backend APIs, model-serving services, evaluation/replay, CI/CD, observability, and incident response. Recent private/open-source work adds hands-on familiarity with the harness around agentic systems: MCP connectors, tool routing, schema validation, approval gates, traceability, replay, and AI-assisted software delivery.

work-experience

Vortexa Ltd, London, UK 12/2020–present — Engineering Manager | ML Systems Lead

Own engineering strategy and delivery for a live post-ingestion ML and streaming estate that turns roughly **6M vessel-position records/hour** into production intelligence for **13.5K monitored vessels**. Lead a **10-person** cross-functional team across engineering, Product, and domain expertise, with **6** direct reports and accountability for platform reliability, product interfaces, model quality, roadmap execution, and delivery maturity.

- **People & Product Delivery:** Manage **6** direct reports; own 1:1s, performance/career development, coaching, onboarding, hiring for **6** roles, team cadence, and delivery alignment with Product, analysts, domain specialists, and engineering stakeholders.
- **Production Platform Ownership:** Lead live backend services and event-driven **Kafka/Flink** pipelines for real-time inference, model serving, signal quality, downstream enrichment, and customer-facing product behavior.
- **Architecture:** Established Flink as a strategic platform direction and led the **Kafka Streams-to-Flink** transformation for critical live pipelines, improving state/checkpointing, operational visibility, maintainability, and compute scaling independent of Kafka partitioning.
- **Model-Serving & MLOps:** Led research-to-production delivery for destination and ETA/waiting-time prediction services; established **MLflow**, model/data versioning, service switch-over, model-serving APIs, replay/evaluation interfaces, live accuracy monitoring, and explainability practices.
- **Quality & Reliability:** Raised engineering quality through CI/CD discipline, local E2E/integration tests, observability, runbooks, rollback/fallback paths, shared on-call, and DORA/Jira delivery signals; maintained **MTTR** under **30** min.
- **Interfaces & Governance:** Drove data lineage, schema-management, Avro/interface-contract, SLA, and domain-owner proposals to make cross-team dependencies explicit and reduce ambiguity across a growing production estate.
- **Engineering Practice:** Standardised I/O seams, repo archetypes, AWS CodeArtifact/shared-library publishing, ADRs, dev containers, Backstage adoption, and local test seams, cutting a **3h** DAG feedback loop to under **5** min.

DataReply, London, UK 04/2016–12/2020 — Senior Consultant | Data Scientist / ML Engineer

Joined Data Reply's London spin-off as its first consultant and helped grow it from **three practitioners and two managers to 30+ consultants**. Progressed to Senior Consultant; shaped client opportunities, agreed deliverables/timelines, managed expectations, placed consultants, interviewed/mentored hires, and delivered data-platform and ML engineering work across enterprise accounts.

- **Vodafone Group:** Led multi-workstream GCP data-science platform work based on Kubeflow; shaped requirements, migration planning, CI/CD, workflow automation, ML lifecycle support, and feature-engineering migration.
- **CNHi:** Lead Data Scientist/Scrum Master on predictive maintenance with Azure/Databricks; translated stakeholder needs into a DS/DE backlog and guided a **PySpark** team over live telemetry.
- **UBS:** Embedded consultant who grew from Data Scientist into ML Engineer through XP/pairing while delivering graph/process-mining analytics, observability, and real-time **Kafka**/Elasticsearch pipelines.
- **Community:** Ran the London Flink Meetup for roughly five years, building a technical community around stream processing, data platforms, and production engineering practice.

KCL, UCL, David Game College, UK 2010–2016 — Teaching and Technical Training

Delivered courses and tutorials in algorithm design, data structures, databases/SQL, and Java/Python during and after doctoral research; developed presentation discipline, learner support, and mentoring habits.

selected agent/platform work

Private	**Promet**	Local AI harness
	Local-first AI workbench for memory-aware, voice-enabled, tool-using assistants; hands-on work on local runtimes, MCP connector setup/discovery, tool routing, schema validation, side-effect policy, approval gates, durable state, OAuth-backed connectors, traces, and replay/evaluation.	
2026	**MCP: Plumbing, Trust**	Public writing
	Practical article on MCP mental models, connector plumbing, staged tool exposure, host-side routing, approval, and evidence boundaries for local/private AI assistants.	
PyPI / JetBrains	**skeleton-replay / IDE plugin**	Runtime evidence
	Public developer tooling that turns script/pytest runs into traces, architecture snapshots, workflow evidence, and replayable IDE reports for review, onboarding, debugging, and LLM-assisted code understanding.	
PyPI	**dynamicio**	Data contracts
	Free public tooling for making I/O seams explicit, validating schema/data expectations, and switching local/dev/prod datasets so pipeline logic can be tested with faster feedback.	

standards/affiliations

since 10/2024	**University College London, Dept. of Information Studies**	Associate Researcher
	Research/teach practical AI standardisation: AI Act, auditability, model/data versioning, explainability, and safe adoption.	
since 01/2021	**ISO - International Organisation for Standardisation**	Committee Expert Member for JTC'21
	Member of JTC'21 Working Group 3, developing AI standards aligned with EU policy and international norms.	

education

2010–2014	**Ph.D. in Computer Science**	King's College London, UK
	***Thesis**: Persuasion Dialogues & Opponent Modelling.*	
	Developed logical inference methods for large **Knowledge Graphs**, combining symbolic reasoning, **graph analytics**, and Bayesian techniques (e.g., MCMC). Received **Best Poster Award** at IJCAI 2013 (413 submissions), **EPSRC PhD Scholarship**, **Outstanding TA Award**, and **Graduate Certificate in Academic Practice**.	
	Access Publications on Google Scholar	
2004–2010	**Diploma (BEng) in Computer Engineering**	University of Thessaly, GR
	Five-year polytechnic degree in Computer Science and Engineering with a **strong focus on mathematics** (10 courses), covering theoretical aspects, networking, and telecommunications. ***Majored in Artificial Intelligence***.	
2002–2004	**Military Officer: Second Lieutenant**	National Guard, CY
	Completed national service as a commissioned Second Lieutenant.	

selected public work

2023	**Agile in Action**	Podcast
	Agile Data Science: Sailing Through Data Science; practical discussion of team practice, delivery, and trustworthy ML systems.	
2022	**Open Data Science Conference**	Talk
	dynamicio: a PyPI library for abstracting I/O in ML/data workflows and creating testable seams around pipeline logic.	